

Event-State Theory VII: Quantum Mechanics as the Statistical Limit of Causal Optimization

Torben Wille

Independent Researcher, Event-State Project

(Dated: January 19, 2026)

Abstract: We derive the formalism of Quantum Mechanics from the axioms of Event-State Theory (EST). While Paper VI established General Relativity as the deterministic limit of computational optimization, this work addresses the stochastic limit where the computational temperature λ is non-negligible. We demonstrate that the "Field of Potential" Ω functions as a phase space of possible next-frames, governed by a Boltzmann-like selection rule. By identifying matter not as point particles but as topological defects ("twists") in the causal graph, we show that the cost function acquires an oscillatory phase component to satisfy lattice continuity. This creates interference effects in the selection probability, reproducing the Feynman Path Integral and the Schrödinger equation. Thus, Quantum Mechanics is identified not as a fundamental framework, but as the statistical mechanics of causal exploration.

I. INTRODUCTION

While the deterministic, large-scale limit of Event-State Theory's cost function gives rise to General Relativity (Paper VI), the microscopic, probabilistic regime reveals a different structure. This paper explores the behavior of the EST selection rule when multiple successor-states have nearly equivalent cost, forcing a probabilistic evolution. We demonstrate that the requirement to represent the EST action unitarily on a discrete graph mathematically necessitates the emergence of complex amplitudes and the Feynman path integral. The Schrödinger Equation is then derived as the continuum limit of this probabilistic causal selection, establishing Quantum Mechanics as the statistical effective theory of the EST framework.

Event-State Theory (EST) takes a different route. It posits a discrete, pregeometric substrate in which the universe is described by a sequence of global event-states $\{\Psi_n\}_{n \in \mathbb{N}}$, ordered causally but not evolving in a pre-given time parameter. Successor states are selected according to the *Path of Causal Least Resistance (PCLR)*, implemented by a computational cost functional

$$J(\Psi_n \rightarrow \Psi_{n+1}) = \alpha K(\Psi_{n+1}|\Psi_n) - \gamma I_F(\Psi_{n+1} \parallel \Psi_n) + \dots, \quad (1)$$

where K is a conditional algorithmic complexity and I_F is an information-fidelity (Fisher-type) measure.¹

The total EST cost functional can be decomposed into a geometric component governing the causal graph itself, and a matter component governing the states on the graph ($J_{\text{Total}} = J_{\text{geom}} + J_{\text{matter}}$). General Relativity emerges when we seek the optimal background geometry by minimizing the total functional ($\delta J_{\text{Total}}/\delta g_{\mu\nu} = 0$). Quantum Mechanics emerges when we analyze the probabilistic dynamics of the matter component within a given

geometry, governed by the statistical weight of its action ($P \propto e^{-J_{\text{matter}}/\lambda}$). This paper focuses on the latter.

In earlier EST work, this structure was used to motivate classical-like dynamics and large-scale structure formation. Here we address a different question: *How does the formal structure of QM—superposition, complex amplitudes, interference, and the Schrödinger equation—arise from the same underlying causal-information principle?*

We proceed in three steps:

1. Show that the PCLR cost functional naturally defines a real, additive "action" $S[H]$ over histories H and induces classical Gibbs weights $\propto \exp(-S/\lambda)$.
2. Prove that any unitary, composition-compatible representation of this additive structure is uniquely realized by a complex phase $\exp(iS/\hbar)$, via the group characters of $(\mathbb{R}, +)$.
3. Show that, in the continuum limit, the resulting amplitude propagation yields the Schrödinger equation and Born rule upon coarse-graining.

In this way, the factor of i in the Feynman weight is not imported from standard quantum theory; it is forced by the requirement that EST's additive cost functional be represented unitarily.

II. PROBABILISTIC CAUSAL SELECTION IN EST

Let $\mathcal{C}_n$ denote the set of admissible successor states to a given global state Ψ_n . EST supplies a cost for each candidate successor $\Phi \in \mathcal{C}_n$ via Eq. (1). When the minimizer of J is unique, evolution is effectively deterministic. However, in generic situations there exist many Φ with

$$J(\Psi_n \rightarrow \Phi) \approx J(\Psi_n \rightarrow \Phi') \approx J_{\min}, \quad (2)$$

so that purely deterministic selection would be unstable under perturbations.

¹ The dots ($\dots$) indicate additional thermodynamic or bias terms, such as 'scar-tissue' asymmetry parameters, which are not central to the present derivation.

A natural and least-biased prescription is to assign a classical transition probability

$$P(\Psi_{n+1}|\Psi_n) = \frac{1}{Z_n} \exp\left[-\frac{1}{\lambda} J(\Psi_n \rightarrow \Psi_{n+1})\right], \quad (3)$$

where λ is an information-scale parameter and Z_n normalizes the distribution.²

A *history* H is a sequence

$$H = (\Psi_0, \Psi_1, \dots, \Psi_N), \quad (4)$$

and its classical probability is

$$P[H] = \prod_{n=0}^{N-1} P(\Psi_{n+1}|\Psi_n) \propto \exp\left[-\frac{1}{\lambda} \sum_{n=0}^{N-1} J(\Psi_n \rightarrow \Psi_{n+1})\right]. \quad (5)$$

We define the *discrete action-like functional*

$$S[H] := \sum_{n=0}^{N-1} J(\Psi_n \rightarrow \Psi_{n+1}), \quad (6)$$

so that

$$P[H] \propto \exp\left(-\frac{S[H]}{\lambda}\right). \quad (7)$$

Crucially, S is *additive* under concatenation of histories: if $H = H_1 \circ H_2$ denotes the concatenation of two partial histories H_1 and H_2 , then

$$S[H_1 \circ H_2] = S[H_1] + S[H_2]. \quad (8)$$

This simple structural property will drive the emergence of complex amplitudes.

III. FROM ADDITIVE ACTION TO COMPLEX AMPLITUDES

At the purely classical level, Eq. (6) and the Gibbs factor $\exp(-S/\lambda)$ already define a stochastic dynamics on histories. Quantum mechanics, however, is characterized by *complex amplitudes* obeying a superposition principle and yielding probabilities via $|\psi|^2$. To recover this structure from EST, we must map the real-valued, additive action $S[H]$ to complex quantities in a way that respects:

1. **Composition:** concatenation of histories corresponds to multiplication of their representations.
2. **Unitarity:** the representation has unit modulus, so total probability is conserved.

3. **Continuity:** small changes in S yield small changes in the representation.

These requirements lead naturally to the notion of a *group character*. Consider the additive group $(\mathbb{R}, +)$ of possible action values. A character is a homomorphism

$$\chi : (\mathbb{R}, +) \rightarrow U(1) \quad (9)$$

satisfying

$$\chi(S_1 + S_2) = \chi(S_1) \chi(S_2), \quad |\chi(S)| = 1, \quad (10)$$

with χ continuous.³

A standard result in harmonic analysis states that all such characters are of the form

$$\chi_k(S) = \exp(ikS), \quad k \in \mathbb{R}. \quad (11)$$

Thus, up to an overall scale factor k , the *only* way to represent the real additive structure of S as a unitary phase factor is through an exponential with an i .

We therefore define an *amplitude map* on histories by

$$\mathcal{A}[H] := \chi(S[H]) = \exp\left(\frac{i}{\hbar} S[H]\right), \quad (12)$$

where we have set $\hbar := 1/k$ as the emergent quantum of action.

Two key properties follow immediately:

1. **Composition:** for concatenated histories $H = H_1 \circ H_2$,

$$\mathcal{A}[H_1 \circ H_2] = \mathcal{A}[H_1] \mathcal{A}[H_2], \quad (13)$$

thanks to Eq. (8).

2. **Norm conservation:** $|\mathcal{A}[H]| = 1$ for each fine-grained history, so probability is conserved under unitary evolution when we form superpositions and take $|\sum_H \mathcal{A}[H]|^2$.

Thus the complex phase factor $\exp(iS/\hbar)$ is not postulated by importing the Feynman rule from external quantum theory; it is *forced* by the requirement that EST's real, additive cost functional be represented by a continuous, unitary, composition-compatible character of the action group.

The classical Gibbs weights $\exp(-S/\lambda)$ and the quantum phases $\exp(iS/\hbar)$ are two complementary uses of the same action-like functional: the former defines a real probability measure over histories, the latter defines their interference structure in the amplitude formalism.

² This is analogous to the maximum-entropy or entropic dynamics principle [3]: given a cost (or “energy”) functional and limited information, the least biased assignment is exponential in that cost.

³ This is a special case of Pontryagin duality: the continuous one-dimensional unitary representations of $(\mathbb{R}, +)$ are in one-to-one correspondence with real numbers k , via $x \mapsto e^{ikx}$.

IV. PATH INTEGRAL AND SCHRÖDINGER EQUATION

Given the amplitude assignment (12), the transition amplitude between coarse-grained configurations (x_i, t_i) and (x_f, t_f) may be written as a sum over histories H connecting them:

$$\mathcal{A}(x_f, t_f; x_i, t_i) = \sum_{H: x_i \rightarrow x_f} \exp\left(\frac{i}{\hbar} S[H]\right). \quad (14)$$

In the continuum limit, this becomes the Feynman path integral [1]

$$\mathcal{A}(x_f, t_f; x_i, t_i) = \int \mathcal{D}x(t) \exp\left[\frac{i}{\hbar} S[x(t)]\right], \quad (15)$$

where $S[x(t)]$ is the continuum action corresponding to the coarse-grained dynamics derived in the classical limit of EST.

Let $\psi(x, t)$ denote the amplitude to find the system at x on the coarse-grained slice labelled by t . For a small time step Δt , amplitude propagation is governed by a kernel K :

$$\psi(x, t + \Delta t) = \int dx' K(x, t + \Delta t; x', t) \psi(x', t), \quad (16)$$

where

$$K(x, t + \Delta t; x', t) = \mathcal{N} \exp\left[\frac{i}{\hbar} S(x, x'; \Delta t)\right]. \quad (17)$$

For a nonrelativistic particle of mass m in potential $V(x)$,

$$S(x, x'; \Delta t) \simeq \frac{m}{2} \frac{(x - x')^2}{\Delta t} - V(x) \Delta t. \quad (18)$$

Performing the Gaussian integral over x' to leading order in Δt and taking the limit $\Delta t \rightarrow 0$ yields the Schrödinger equation [1]:

$$i\hbar \frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m} \nabla^2 \psi + V(x) \psi. \quad (19)$$

Thus the Schrödinger dynamics emerges as the continuum limit of amplitude propagation induced by EST's discrete causal selection rule, once we represent the action as a unitary character.

V. DECOHERENCE AND CLASSICALITY

In EST, the microscopic causal graph contains a vast number of fine-grained histories. Macroscopic observations correspond to coarse-grained descriptions in which many microscopically distinct histories are indistinguishable. Let $\{\alpha\}$ be a set of coarse-grained alternatives (e.g., ranges of positions) and let $\mathcal{H}_\alpha$ denote the set of fine-grained histories realizing α .

The amplitude associated with α is

$$\mathcal{A}_\alpha = \sum_{H \in \mathcal{H}_\alpha} \mathcal{A}[H]. \quad (20)$$

The corresponding probability is

$$P(\alpha) = |\mathcal{A}_\alpha|^2 = \sum_{H, H' \in \mathcal{H}_\alpha} \exp\left[\frac{i}{\hbar} (S[H] - S[H'])\right]. \quad (21)$$

When environmental interactions or coarse-graining suppress interference terms with $H \neq H'$, the off-diagonal contributions vanish and we obtain an effectively classical probability distribution over α :

$$P(\alpha) \approx \sum_{H \in \mathcal{H}_\alpha} 1 \propto |\mathcal{A}_\alpha|^2. \quad (22)$$

This is decoherence in the sense of Zurek [4]: classicality emerges as a consequence of information loss (coarse-graining) over fine-grained histories.⁴

VI. BORN RULE FROM CAUSAL MULTIPLICITY

The Born rule,

$$P(x) = |\psi(x)|^2, \quad (23)$$

is often taken as a primitive axiom of QM. In EST, it emerges from the combinatorial multiplicity of histories converging on a given coarse-grained configuration.

Let $\psi(x)$ be the coarse-grained amplitude

$$\psi(x) = \sum_{H \in \mathcal{H}(x)} \mathcal{A}[H], \quad (24)$$

where $\mathcal{H}(x)$ is the set of fine-grained histories that arrive at x . Then

$$P(x) = |\psi(x)|^2 = \sum_{H, H' \in \mathcal{H}(x)} \exp\left[\frac{i}{\hbar} (S[H] - S[H'])\right]. \quad (25)$$

Under decoherence or coarse-graining, the cross terms with $H \neq H'$ tend to cancel or average out, leaving

$$P(x) \approx \sum_{H \in \mathcal{H}(x)} 1 \propto |\psi(x)|^2. \quad (26)$$

Thus the Born rule arises from the squared modulus of the sum over amplitudes, reflecting the counting of effectively indistinguishable causal histories.

⁴ In EST, decoherence has a particularly concrete meaning: many microscopically distinct causal graphs become indistinguishable at the macroscopic scale, washing out phase relations between them.

VII. EMERGENT PARTICLES AND FIELDS

Within EST, “particles” are stable, localized patterns of causal propagation, i.e. recurrent motifs in the event-state graph that preserve their identity through many updates. “Fields” correspond to collective excitations of the underlying causal substrate, described in the continuum limit by smooth field configurations. Quantization arises from the same amplitude assignment (12) applied to field histories, leading to the usual quantum field theoretic structures.

Local commutation relations can be traced back to the underlying causal structure: spacelike-separated events have no causal links and thus their update operators commute, reflecting microcausality in the emergent field description.

VIII. PREDICTIONS AND FALSIFIABILITY

Because QM emerges from a discrete causal-information substrate, EST predicts potential deviations from standard QM in regimes where:

1. The number of contributing fine-grained histories is small (low causal multiplicity).
2. Microscopic anisotropies in the causal graph are large, leading to small violations of Lorentz symmetry at very high energies.
3. Coarse-graining is incomplete, producing residual phase correlations that could manifest as anomalous interference patterns.

4. Pre-causal resonance (PCR) effects may appear in high-energy collider data, as subtle correlations in vacuum fluctuations preceding nominal “trigger” events.

These possibilities distinguish EST from purely axiomatic reconstructions of QM that do not posit an underlying discrete causal substrate.

IX. CONCLUSION

We have shown how the formal structure of quantum mechanics can be derived from the successor-state selection rule of Event-State Theory. EST provides a real, additive action-like functional over histories, $S[H]$, from its causal optimization cost J . Classical probabilistic dynamics arises from the Gibbs weights $\exp(-S/\lambda)$, while the uniquely determined unitary character $\exp(iS/\hbar)$ of the additive action group yields complex amplitudes. The Feynman path integral, Schrödinger equation, and Born rule then follow as the continuum and coarse-grained limits of this discrete causal-information dynamics.

Quantum mechanics, in this view, is not an independent set of axioms but the effective theory describing how macroscopic observers sample and coarse-grain the vast space of microscopic causal histories selected by the Path of Causal Least Resistance.

-
- [1] R. P. Feynman, “Space-Time Approach to Non-Relativistic Quantum Mechanics,” *Rev. Mod. Phys.* **20**, 367 (1948).
 - [2] M. Gell-Mann and J. Hartle, “Classical Equations for Quantum Systems,” *Phys. Rev. D* **47**, 3345 (1993).
 - [3] A. Caticha, “Entropic Dynamics, Time and Quantum Theory,” *J. Phys. A* **44**, 225303 (2011).
 - [4] W. H. Zurek, “Decoherence, einselection, and the quantum origins of the classical,” *Rev. Mod. Phys.* **75**, 715 (2003).
 - [5] Wille, T. (2025). *EST I: Theoretical Framework*. Zenodo. <https://doi.org/10.5281/zenodo.17698703>
 - [6] Wille, T. (2025). *EST II: Computational Proof*. Zenodo. <https://doi.org/10.5281/zenodo.17769476>
 - [7] Wille, T. (2025). *EST III: Computational Synthesis*. Zenodo. <https://doi.org/10.5281/zenodo.17798980>
 - [8] Wille, T. (2025). *EST IV: Statistical Homology*. <https://doi.org/10.5281/zenodo.17877635>
 - [9] Wille, T. (2026). *EST VI: General Relativity as the Infrared Effective Theory of Discrete Causal Optimization*.
 - [10] Wille, T. (2026). *EST IX: A Programmatic Bridge Between Variational Classical Physics and Unitary Quantum Interference*. In preparation. <https://doi.org/10.5281/zenodo.18219699>